

LOW ACRYLONITRILE NBR: PATENT LANDSCAPE AND SYNTHESIS TRENDS

1. ABSTRACT

This report reviews patent evidence relevant to low-acrylonitrile nitrile butadiene rubber (NBR), with emphasis on how acrylonitrile content is controlled through monomer feed, solvent choice, initiator/modifier selection, and polymerization mode. The single primary patent in the manifest, US9023914B2, is not a dedicated low-ACN grade patent; instead, it discloses solution free-radical polymerization of NBR in organic solvents using specific modifier compounds to tune molecular architecture, molecular weight, and polydispersity. Within its examples, the patent demonstrates NBR synthesis from acrylonitrile and butadiene at fixed monomer ratios, including a representative formulation of 37 phm acrylonitrile and 63 phm butadiene, polymerized at 100°C under 3 bar butadiene pressure, as well as a lower-acrylonitrile variant using 30 phm acrylonitrile and 50 phm butadiene. The disclosed process is notable for its solution-phase operation, oxygen exclusion, use of azo initiator and RAFT-type modifier chemistry, and precipitation isolation from alcohol or methanol. Although the patent's main objective is not low-ACN composition per se, it is technically relevant because it provides a controllable organic-solution route to NBR with explicit monomer-feed adjustment and molecular-weight control, both of which are central to designing lower-acrylonitrile grades. The broader process trend reflected here is a move away from conventional emulsion polymerization toward solution polymerization for improved control over architecture and downstream compatibility, especially where subsequent hydrogenation or tailored performance is desired. For low-ACN NBR synthesis, the evidence supports using acrylonitrile/butadiene feed adjustment in solution, inert handling, moderate-to-high polymerization temperature, and modifier-assisted control to achieve targeted composition and viscosity.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US9023914B2
- Patent Title: Nitrile rubbers and production thereof in organic solvents
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2015
- Legal Status: Unknown

- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses free-radical solution polymerization of acrylonitrile and 1,3-butadiene in an organic solvent in the presence of specific modifier compounds, with optional subsequent hydrogenation. The process is carried out under oxygen-free conditions and uses precipitation to isolate the nitrile rubber after polymerization. - Acrylonitrile: 43 ml (653.3 mmol, 37 phm) — Example 1 - 1,3-Butadiene: 90 ml (1077 mmol, 63 phm) — Example 1 - Dimethylacetamide: 50 ml (51 phm) — Example 1 - Vazo® 88: 284.9 mg (1.17 mmol, 0.31 phm) — Example 1 - DoPAT: 204 mg (0.583 mmol, 0.22 phm) — Example 1 - Butadiene pressure: 3 bar — Example 1 - Reaction temperature: 100°C — Example 1 - Polymerization time: 9 hours — Example 1 - Precipitation solvent: ethanol containing pyrocatechol/hydroquinone stabilizer — Example 1 - Hydroxyethyl acrylate: 20 phm — Table 1, Example 6 - Acrylonitrile: 30 phm — Table 1, Example 6 - 1,3-Butadiene: 50 phm — Table 1, Example 6 - Polymerization time: 6 h — Example 6 - Final conversion: 39% — Example 6 - Modifier concentration: 2.22 phm — Table 1, Example 7 - Polymerization time: 9 h — Example 7 - Final conversion: 48% — Example 7 - NBR #7 Mn: 12 000 g/mol — Example 8 - Styrene: 2.79 g (26.8 mmol, 50 phm) — Example 8 - DMAc: 2 g — Example 8 - Reaction temperature: 100°C — Example 8 - Polymerization time: 13 hours — Example 8 - Final conversion: 9.2% — Example 8 - Pyrocatechol/hydroquinone: 5.7 phm — Table 1, Examples 1-8 - Polymerization temperature: 100°C — Table 1, Examples 1-8 - Polymerization time: 5 h — Table 1, Example 5 - Final conversion: 28% — Table 1, Example 5 - Final conversion: 50% — Table 1, Example 1 - Final conversion: 49% — Table 1, Example 2 - Final conversion: 51% — Table 1, Example 3 - Final conversion: 53% — Table 1, Example 4

Relevant Experimental Evidence - Example 1: Demonstrated solution polymerization of NBR from 37 phm acrylonitrile and 63 phm butadiene at 100°C and 3 bar, yielding polymer after 9 h and precipitation from ethanol. - Example 2: Showed that changing modifier and initiator amounts while keeping the same monomer ratio still gave NBR under the same 100°C solution conditions. - Example 3: Demonstrated that an alternative solvent system could be used while maintaining solution polymerization of NBR. - Example 6: Demonstrated a lower-acrylonitrile terpolymer formulation with 30 phm acrylonitrile and 50 phm butadiene plus 20 phm hydroxyethyl acrylate, terminated after 6 h at 39% conversion. - Example 7: Demonstrated deliberate preparation of a low-molecular-mass NBR by increasing modifier loading to 2.22 phm, reaching 48% conversion after 9 h.

Technical Relevance This patent is relevant because it provides a solution-polymerization platform where acrylonitrile feed can be adjusted directly, including a lower-ACN example. Its modifier-controlled process and explicit monomer ratios make it useful for designing low-acrylonitrile NBR grades with controlled molecular weight and composition.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- No qualifying primary patents were identified under the configured target material relevance criteria. Cross-patent quantitative trends were therefore not generated.

4. CONCLUSION

US9023914B2 shows that low-acrylonitrile NBR can be approached through organic-solution free-radical copolymerization with explicit monomer-feed control, inert handling, and modifier-assisted molecular-weight regulation. The most directly useful evidence is the lower-ACN example using 30 phm acrylonitrile and 50 phm butadiene at 100°C, with precipitation isolation after polymerization. For low-ACN NBR synthesis, the evidence supports solution polymerization at about 100°C, careful oxygen exclusion, and use of modifier/initiator systems to maintain processability while tuning composition and viscosity.

5. REFERENCES

- US9023914B2 | Nitrile rubbers and production thereof in organic solvents | Not disclosed | US | 2015 | <https://patents.google.com/patent/US9023914B2/en>